C# BACKEND TRACK | OOP PHASE

Task 1

System Analysis Task

Think about the business. Extract the models. Assign the sources.

Codeline Bootcamp · Backend Phase · OOP Session 3 Task

SOURCE TAG LEGEND

Every property of every class model gets its value from somewhere. Before writing any code, you must decide **where** each value comes from. Use one of these five source tags to label each property in your analysis:

System Generated	**System Generated**	The system assigns this value automatically. The user never types it. Example: an auto-incremented ID calculated as List.Count + 1.
User Input	**User Input**	The user types this value directly at the keyboard when prompted by the system.
Default Value	**Default Value**	A fixed value set by the code when the object is created. The user never chooses it. Example: isBooked = false, status = "Active".
Calculated	**Calculated**	Derived automatically from another object already in memory — no user prompt. Example: appointment date copied from a chosen slot, fee copied from a doctor.
From List	**From List**	The user selects a value from a list of options displayed on screen. Example: choosing a doctor ID from a printed list of doctors.

Domain 1 — Library Management System

Extract everything yourself

Read the following description of a real library carefully. Your job is to go through the 6-phase process and fill in all the blank tables below. Nothing has been done for you — you identify the entities, you decide the properties, and you assign the source tag for each one.

> A public library allows community members to register and borrow books. Each book has a title, author, genre, and a fixed number of physical copies in stock. A registered member can borrow an available book copy and must return it within 14 days. If the book is returned late, the system calculates a fine based on how many days overdue it is. A member can also reserve a book that is currently fully borrowed so they are notified when it becomes available. The librarian needs to see all currently active loans, view all overdue loans sorted by how late they are, and get a report of the most-borrowed books ranked by total borrow count.

Step 1 — Identify the Entities (Phases 1 & 2)

From the description above, identify the main real-world objects (classes) the system must track. Write each entity name, its properties, the data type for each, and the source tag that describes how each property gets its value.

Entity Name: Books

Property Name	Data Type	Source Tag
BookID	int	System Generated
Title	string	User Input
Author	string	User Input
Genre	string	User Input
No.PhysicalCopies	int	Default Value
Total.CountBrrow	int	Default Value
Availibility	bool	Default Value

Entity Name: Member

Property Name	Data Type	Source Tag
MemberID	int	System Generated
MemberName	string	User Input
Phone.NO	int	User Input
Email	string	User Input
Gender	string	User Input
Membership.Date	String	System Generated
Actives.Loans	bool	Default Value

Entity Name: Loans

Property Name	Data Type	Source Tag
LoansID	int	System Generated
MemberID	int	From List
BookID	int	From List
BorrowDate	string	System Generated
DueDate	string	calculated(within 14 days)
ReturnDate	string	User input
FineAmount	decimal	calculated

Entity Name: Librarian

Property Name	Data Type	Source Tag
LibrarianID	int	System Generated
LibrarianName	string	User input
Email	string	User input
PhoneNumber	int	User input

If you identify more than 4 entities, continue on the back of this sheet or add extra rows above.

Step 2 — Describe the Entity Relationships (Phase 1 continued)

How do the entities connect to each other? Describe the full cycle in plain English — who does what, which entity gets created, and what changes in other entities as a result.

Write the relationship cycle here:

Member (1) → (M) Loan
One member can make many loans.
Each loan belongs to only one member.

Librarian (1) → (M) Loan
One librarian can handle many loan transactions.
Each loan is processed by one librarian.

Book (1) → (M) Loan
One book can appear in many loan records over time.
Each loan is for one specific book.

Step 3 — List the Services (Phase 4)

From the description, identify every service (function) the system needs. Write at least 8 services — give each a name and a one-line description of what it does.

#	Service Name	What it does
1	RegisterMember	Creates a new library member
2	AddBook	Adds a new book to the library
3	ViewBooksByGenre	Displays all books with same Genre
4	BorrowBook	Creates a loan record for a member
5	ReturnBook	Processes returned books
6	CalculateFine	Calculates overdue fines
7	NotifyMember	Notifies a member when a reserved book becomes available
8	ViewActiveLoans	Shows all active loans
9	ViewOverdueLoans	Displays overdue loans sorted by overdue days
10	GenerateMostBorrowedBooksReport	Displays books ranked by borrow count

Domain 2 — Hotel Management System

Assign the source tags

The class models and all their properties are already defined for you below. Your job is to **fill in the Source Tag column** for every single property. Use the five source tags from the legend on page 1: System Generated, User Input, Default Value, Calculated, From List.

A hotel allows guests to register and book rooms for a stay. The hotel has different room types (Single, Double, Suite) each with its own nightly price. Each room can only have one active booking at a time. When a guest checks in, a check-in record is created. The total price of a stay is calculated from the number of nights and the room's nightly rate. When a guest checks out, the room becomes available again. A booking can also be cancelled before check-in, which frees the room immediately.

Your task: For each property in the tables below, write the correct source tag in the empty column on the right. Think carefully — ask yourself: does the user type this? does the system generate it? is it copied from another object? does it have a fixed starting value? does the user pick it from a list?

Guest

Property Name	Data Type	Source Tag (fill in)
`guestId`	int	System Generated
`guestName`	string	User input
`guestPhone`	string	User input
`guestEmail`	string	User input
`guestNationality`	string	User input
`registrationDate`	string	System Generated

Room

Property Name	Data Type	Source Tag (fill in)
`roomId`	`int`	System Generated
`roomNumber`	`string`	System Generated
`roomType`	`string`	User Input
`floorNumber`	`int`	User Input
`nightlyPrice`	`decimal`	Calculated
`isAvailable`	`bool`	Default Value

Booking

Property Name	Data Type	Source Tag (fill in)
`bookingId`	`int`	System Generated
`guestId`	`int`	From list
`roomId`	`int`	From list
`checkInDate`	`string`	User input
`checkOutDate`	`string`	User input
`numberOfNights`	`int`	calculated
`totalPrice`	`decimal`	calculated
`status`	`string`	Default Value

CheckInRecord		
Property Name	**Data Type**	**Source Tag (fill in)**
`recordId`	`int`	System Generated
`bookingId`	`int`	From list
`guestId`	`int`	From list
`roomId`	`int`	From list
`actualCheckIn`	`string`	System Generated
`actualCheckOut`	`string`	System Generated
`finalAmount`	`decimal`	calculated

Step 2 — Describe the Entity Relationships

Now that you have studied the four class models, describe in your own words how they connect to each other. Who creates what? What changes when a booking is made? What happens when a guest checks out?

Write the relationship cycle here:

Guest (1) → (M) Booking
One guest can create many bookings.
Each booking belongs to one guest.

Room (1) → (M) Booking
One room can have many bookings over time.
Each booking is assigned to one room.
(Only one active booking at a time per room.)

Booking (1) → (1..1) CheckInRecord
A booking may create a check-in record when the guest arrives.
Each check-in record belongs to exactly one booking.

Step 3 — List the Services

Based on the description and the class models above, list all the services (functions) this hotel system needs. Give each a name and a one-line description.

#	Service Name	What it does
1	RegisterGuest	Registers a new guest
2	AddRoom	Adds a room to the hotel
3	ViewAvailableRooms	Shows available rooms
4	CreateBooking	Creates a room booking
5	CancelBooking	Cancels an existing booking
6	CalculateStayCost	Calculates total booking cost
7	CheckInGuest	Creates a check-in record

#	Service Name	What it does
8	CheckOutGuest	Completes guest checkout
9	ViewBookings	Displays all bookings